EMPLOYEE DETAILS AND PAYROLL MANAGEMENT SYSTEM

Integrated OOPS and DBMS project report

Submitted by

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of

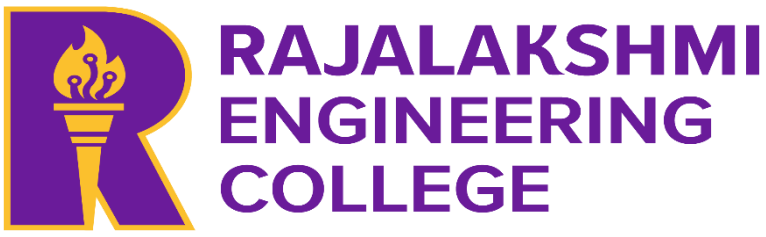
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BONAFIDE CERTIFICATE

Certified that this project report titled "**EMPLOYEE DETAILS AND PAYROLL MANAGEMENT SYSTEM**" is the bonafide work of "**KAARUNYA SHREE G (221001065), LAVANYA R (221001081), KARTHICKARJUN K (221001071)**" who carried out the project work under my supervision.

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ABSTRACT

In today's world the management system is very important and essential for every system. This management system is an application-based system, to manage employees to mark their details. Every organisation whether the government or private uses an information system to store data of their staff. This project is useful for easy user interface. The system uses the powerful database management system, data retrieval and data manipulation. This project provides more ease for managing the data than manually maintaining the data. Hence it saves the lot of time of ours also. So ,we can say that the project is useful for saving valuable time and reducing huge paper work.The main objective of this mini project is to add, update, delete, and view the student details.The Employee Management System is a comprehensive solution designed to streamline and automate various aspects of employee administration within an organization. This mini-project aims to develop a user-friendly and efficient system that facilitates the management of employee information, attendance, leave, and performance evaluation. Employee management system capture and store detailed employee information including the personal details, contact information, and job-related details. This project helps us in organizing the data that needs to be accessed. The System was developed using basic technologies such as MYSQL database and Java (JDBC).

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1 - INTRODUCTION

1.1 PROBLEM STATEMENT

Employee details management system to automate various aspects of employee administration within an organization.

1.2 PROJECT OVERVIEW

The Employee Management System is a comprehensive solution designed to streamline and automate various aspects of employee administration within an organization. This mini-project aims to develop a user-friendly and efficient system that facilitates the management of employee information, attendance, leave, and performance evaluation. Employee management system capture and store detailed employee information including the personal details, contact information, and job-related details. This project helps us in organizing the data that needs to be accessed.

1.3 PURPOSE

The system is aimed at total user-friendly as well as efficient management of varied tasks. To centralize and organize employee information in a systematic manner and to facilitate easy access to accurate and up-to-date employee data, reducing manual errors and ensuring data integrity. To automate the tracking of employee attendance. It enables timely and accurate recording of working hours, aiding in payroll processing and identifying attendance trends. It improves the transparency, reduces paperwork, and ensures consistent application of leave policies. To provide a structured system for accessing and documenting employee performance. To safeguard the sensitive employee information. To allow for potential expansion and inclusion of additional features. It ensures the system can adapt to the evolving needs of the organisation.

1.4 SCOPE

The system is aimed at total user-friendly as well as efficient management of different tasks. They may range from capturing and storing the essential employee information, including personal details, contact information and job-related data. To develop a system for tracking employee attendance through clock-in and clock-out functionalities. To generate basic attendance reports for administrators. To develop basic integration with payroll functionalities, automating salary calculations based on attendance and other relevant factors. To design an intuitive and user-friendly interface for administrators and employees.

2 - FUNCTIONALITIES PROVIDED

FUNCTIONALITIES PROVIDED BY EMPLOYEE MANAGEMENT SYSTEM:

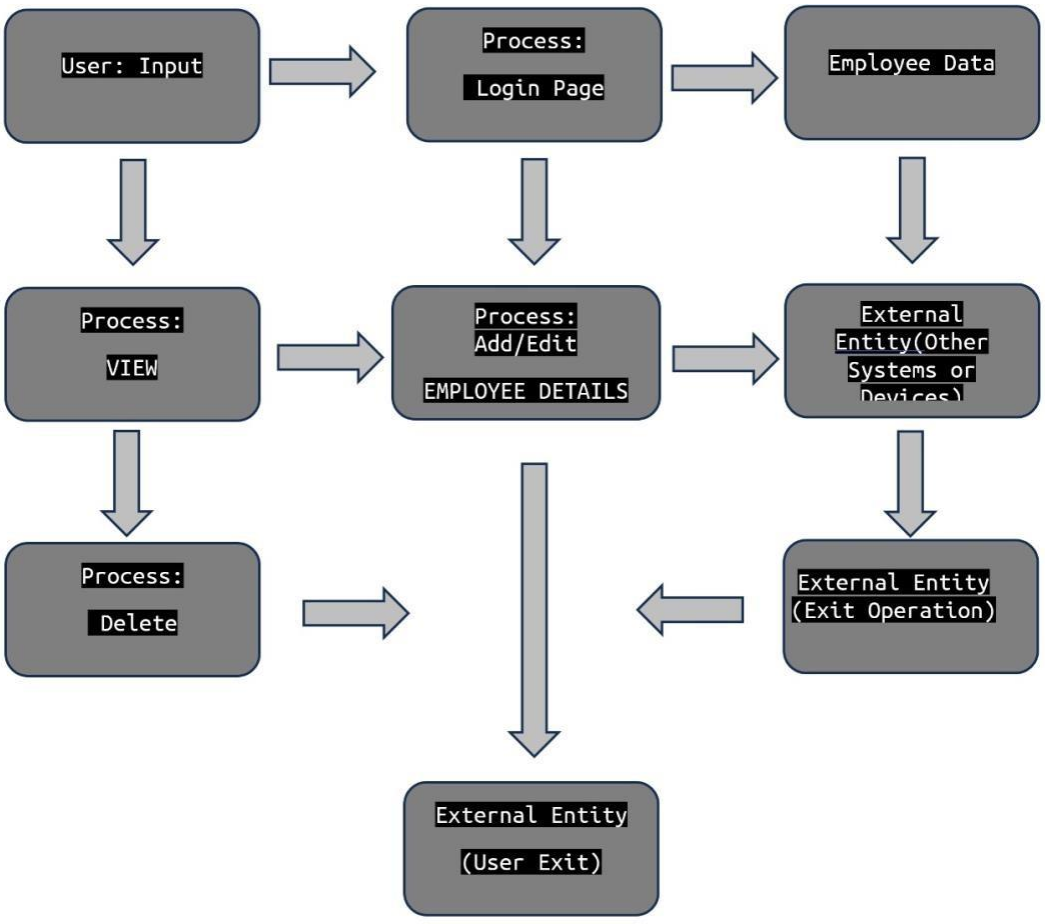
- Simplifying and streamlining all tasks.
- Provide timely communication and remainders.
- Easy to understand.
- Easy to access.
- Easy to handle.
- Facilitate structured performance assessments.
- Automate payroll calculations.

3 - SYSTEM FLOW DIAGRAM

System flow diagram helps in understanding the flow of information and processes within the system. The below displays such a representation for the project on Employee details and payroll management system.

3.1 DATAFLOW DIAGRAM

A Data Flow Diagram (DFD) is a visual way to show how data moves through a system or process. The below displays such a representation for the project on Employee details and payroll management system.



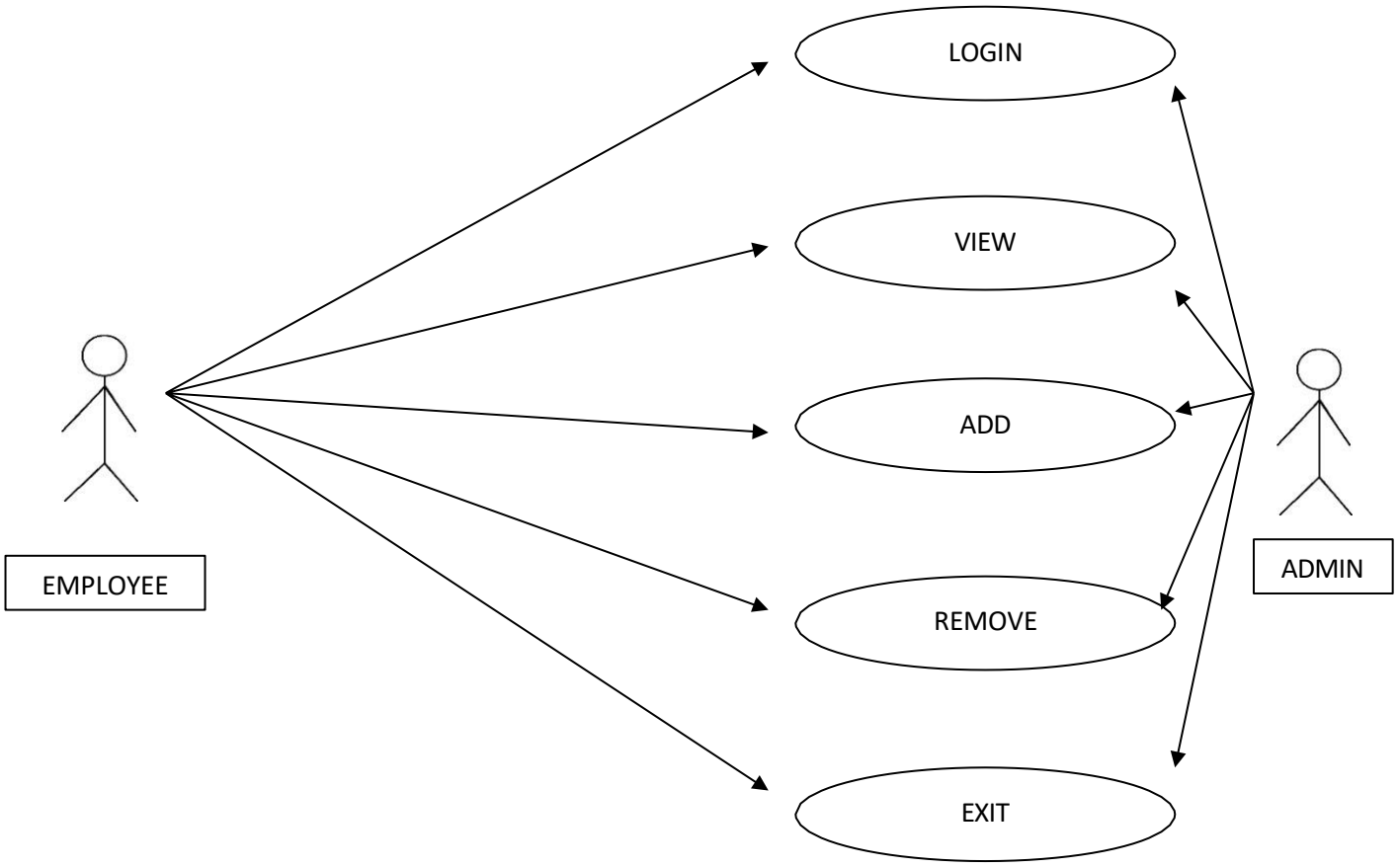
3.2 USE CASE DIAGRAM

A use case diagram is a visual representation that depicts the interactions between actors (users) and a system, illustrating various ways the system can be utilized or the functionalities it offers. The below displays such a representation for the project on Employee details and payroll management system.

Actor: Employee

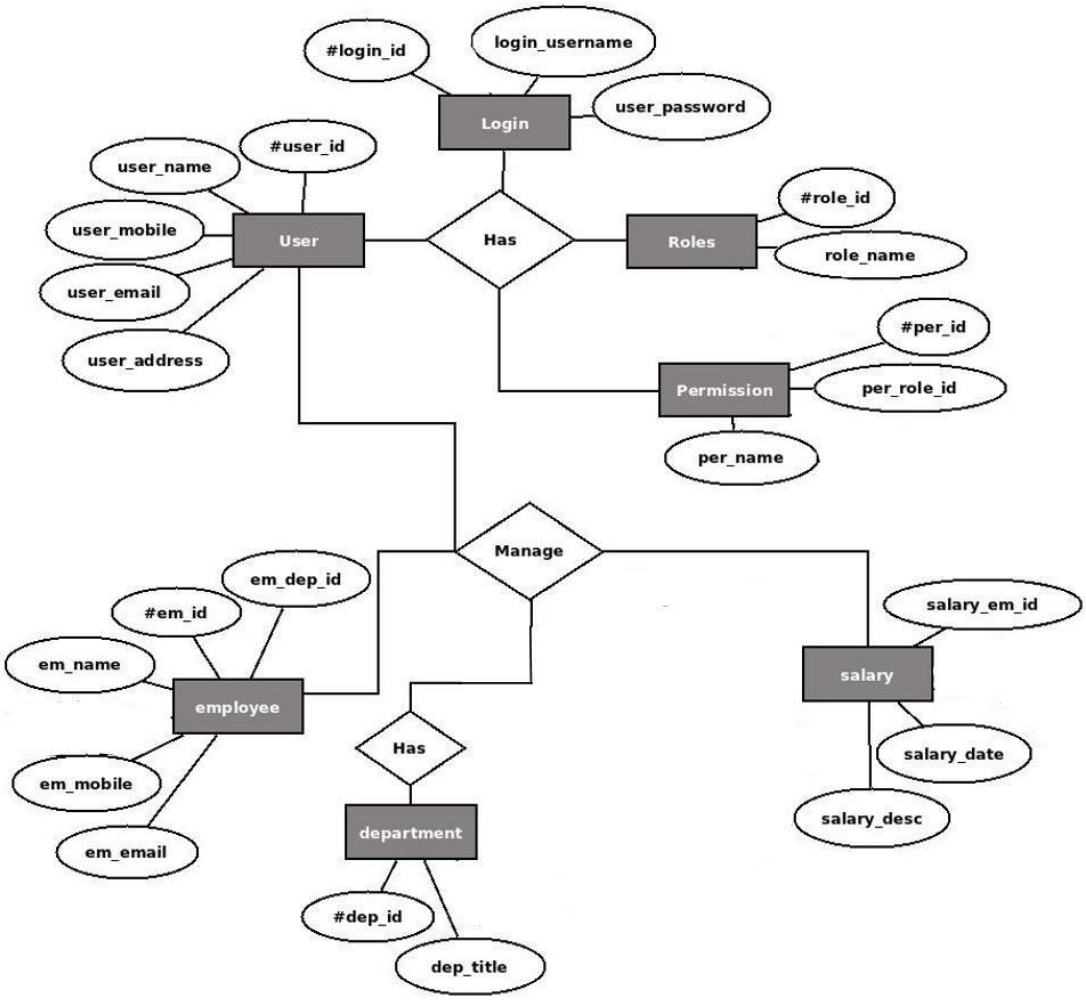
Use Cases

a. Add Employee - Description: Allows an employee to add new employee details to the system.
b. Remove Employee - Description: Permits an employee to remove an existing employee from the system.
c. View Employee Details - Description: Enables an employee to view details of other employees.
d. Exit System - Description: Allows an employee to log out or exit the system.



3.3 ENTITY RELATIONSHIP DIAGRAM

An entity-relationship diagram (ERD) is a visual representation illustrating the relationships between different entities within a database system. The below displays such a representation for the project on Employee details and payroll management system.



4 - MODULE DESCRIPTION

The different types of modules present in this project are:

SPLASH PAGE:

In this section a splash of the intro page is popped up and it may also function as a landing page to attract visitors.

ADD EMPLOYEE:

In this section admin can add employee details. Get data from user and add a employee to the list of employees. While adding the employee into the list, check for the uniqueness of the id number.

VIEW EMPLOYEE:

In this section admin can view and update employee details. This function is to find all the employees who have added data.

DELETE EMPLOYEE:

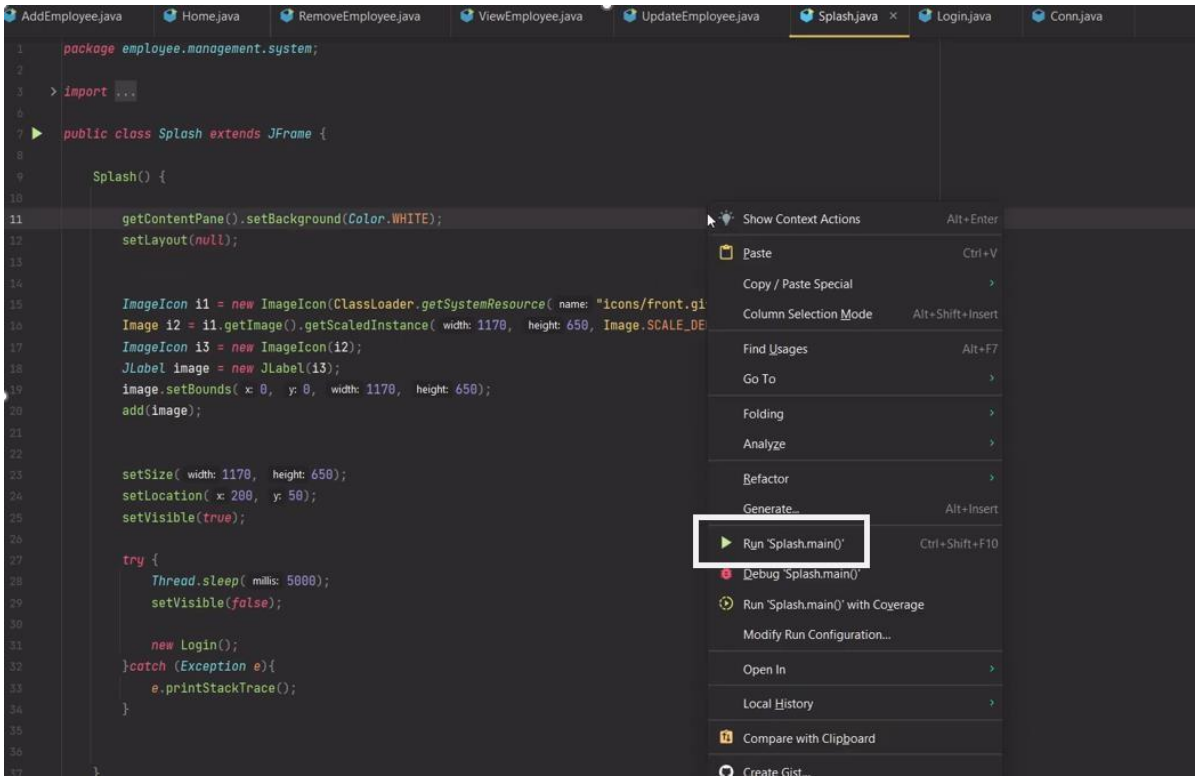
In this section admin can delete employee details.

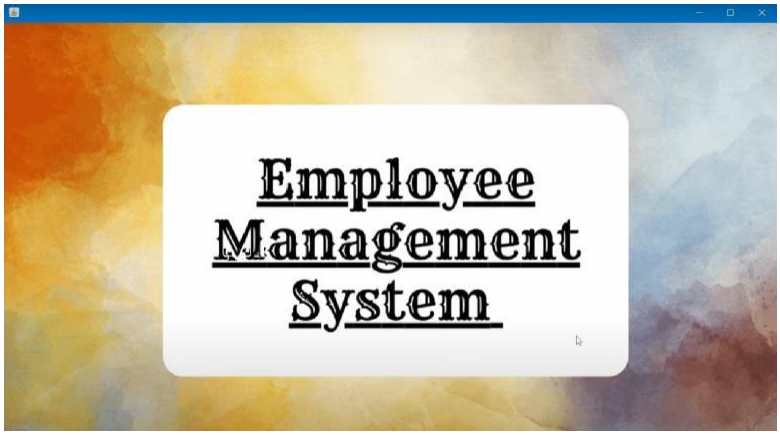
UPDATE EMPLOYEE DETAILS:

In this section admin can update employee details. This function is to update the employee records. This function does not ask for new details for all fields but the user should be able to pick and choose what he wants to update.

5 - IMPLEMENTATION

5.1 SPLASH PAGE :





JAVA CODE:

```
package employee.management.system;

import javax.swing.*;
import java.awt.*;

public class Splash extends JFrame {

    Splash(){

        ImageIcon i1 = new ImageIcon(ClassLoader.getSystemResource("icons/front.gif"));
        Image i2 = i1.getImage().getScaledInstance(1170,650, Image.SCALE_DEFAULT);
        ImageIcon i3 = new ImageIcon(i2);
        JLabel image = new JLabel(i3);
        image.setBounds(0,0,1170,650);
        add(image);


        setSize(1170,650);
        setLocation(200,50);
        setLayout(null);
        setVisible(true);

        try{
            Thread.sleep(5000);
            setVisible(false);
            new Login();
        }catch (Exception e){
            e.printStackTrace();
        }

    }

    public static void main(String[] args) {
        new Splash();
    }
}
```

JDBC CONNECTION CODE:

```
package employee.management.system;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;

public class conn {
```

```
    Connection connection;
    Statement statement;

    public conn(){
        try{
            Class.forName("com.mysql.cj.jdbc.Driver");
            connection =
DriverManager.getConnection("jdbc:mysql://localhost:3306/employeemanagement","root","AyushVish")
;
            statement = connection.createStatement();

        }catch (Exception e){
            e.printStackTrace();
        }
    } }
```

JAVA MAIN CLASS CODE:

```
package employee.management.system;

import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class Main_class extends JFrame {
    Main_class(){

        ImageIcon i1 = new ImageIcon(ClassLoader.getSystemResource("icons/home.jpg"));
        Image i2 = i1.getImage().getScaledInstance(1120,630,Image.SCALE_DEFAULT);
        ImageIcon i3 = new ImageIcon(i2);
        JLabel img = new JLabel(i3);
        img.setBounds(0,0,1120,630);
        add(img);

        JLabel heading = new JLabel("Employee Management System");
        heading.setBounds(340,155,400,40);
        heading.setFont(new Font("Raleway",Font.BOLD,25));
        img.add(heading);

        JButton add = new JButton("Add Employee");
        add.setBounds(335,270,150,40);
        add.setForeground(Color.WHITE);
        add.setBackground(Color.black);
        add.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                new AddEmployee();
                setVisible(false);
            }
        });
        img.add(add);

        JButton view = new JButton("View Employee");
        view.setBounds(565,270,150,40);
        view.setForeground(Color.WHITE);
        view.setBackground(Color.black);
        view.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                new View_Employee();
                setVisible(false);
            }
        });
        img.add(view);

        JButton rem = new JButton("Remove Employee");
```

```
        rem.setBounds(440,370,150,40);
        rem.setForeground(Color.WHITE);
        rem.setBackground(Color.black);
        rem.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                new RemoveEmployee();
            }
        });
        img.add(rem);

        setSize(1120,630);
        setLocation(250,100);
        setLayout(null);
        setVisible(true);

    }
    public static void main(String[] args) {
        new Main_class();
    }
}
```

5.2 LOGIN PAGE :




```
package employee.management.system;

import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.sql.ResultSet;

public class Login extends JFrame implements ActionListener {

    JTextField tusername;
    JPasswordField tpassword;
    JButton login, back;

    Login(){

        JLabel username = new JLabel("Username");
        username.setBounds(40,20,100,30);
        add(username);

        tusername = new JTextField();
        tusername.setBounds(150,20,150,30);
        add(tusername);

        JLabel password = new JLabel("Password");
        password.setBounds(40,70,100,30);
        add(password);

        tpassword = new JPasswordField();
        tpassword.setBounds(150,70,150,30);
        add(tpassword);

        login = new JButton("LOGIN");
```

```
        login.setBounds(150,140,150,30);
        login.setBackground(Color.black);
        login.setForeground(Color.WHITE);
        login.addActionListener(this);
        add(login);

        back = new JButton("BACK");
        back.setBounds(150,180,150,30);
        back.setBackground(Color.black);
        back.setForeground(Color.WHITE);
        back.addActionListener(this);
        add(back);

        ImageIcon i11 = new ImageIcon(ClassLoader.getSystemResource("icons/second.jpg"));
        Image i22 = i11.getImage().getScaledInstance(600,400,Image.SCALE_DEFAULT);
        ImageIcon i33 = new ImageIcon(i22);
        JLabel imgg = new JLabel(i33);
        imgg.setBounds(350,10,600,400);
        add(imgg);


        ImageIcon i1 = new ImageIcon(ClassLoader.getSystemResource("icons/LoginB.jpg"));
        Image i2 = i1.getImage().getScaledInstance(600,300,Image.SCALE_DEFAULT);
        ImageIcon i3 = new ImageIcon(i2);
        JLabel img = new JLabel(i3);
        img.setBounds(0,0,600,300);
        add(img);

        setSize(600,300);
        setLocation(450,200);
        setLayout(null);
        setVisible(true);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == login){
            try {
                String username = tusername.getText();
                String password = tpassword.getText();

                conn conn = new conn();
                String query = "select * from login where username = '"+ username +"' and password =
'"+password+"'";
                ResultSet resultSet = conn.statement.executeQuery(query);
                if (resultSet.next()){
                    setVisible(false);
                    new Main_class();
                }else {
                    JOptionPane.showMessageDialog(null,"Invalid username or password");
                }
```

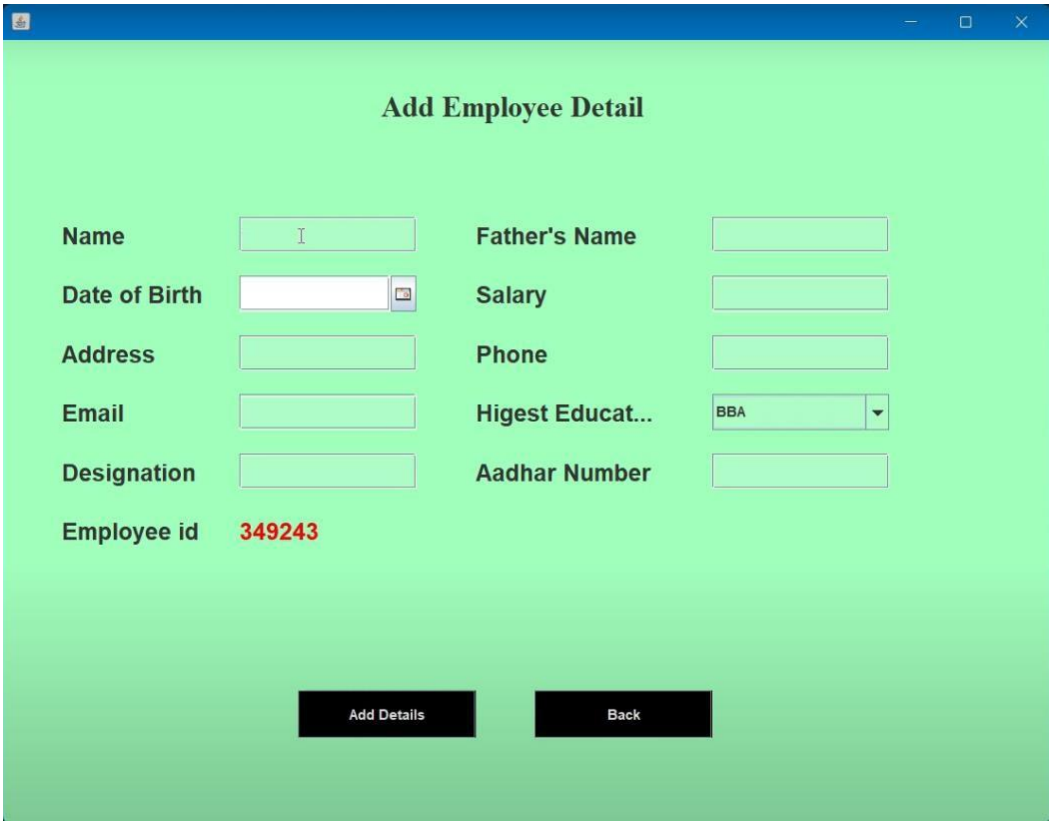
```
            }catch (Exception E){
                E.printStackTrace();
            }

        } else if (e.getSource() == back) {
            System.exit(90);
        }
    }

    public static void main(String[] args) {
        new Login();
    }
}
```

5.3 ADD EMPLOYEE:








```
package employee.management.system;

import com.toedter.calendar.JDateChooser;

import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.util.Random;

public class AddEmployee extends JFrame implements ActionListener {
    Random ran = new Random();
    int number = ran.nextInt(999999);
    JTextField tname, tfname, taddress,tphone, taadhar, temail, tsalary,tdesignation;
    JLabel tempid;
    JDateChooser tdob;

    JButton add,back;

    JComboBox Boxeducation;
    AddEmployee(){

        getContentPane().setBackground(new Color(163,255,188));

        JLabel heading = new JLabel("Add Employee Detail");
        heading.setBounds(320,30,500,50);
        heading.setFont(new Font("serif", Font.BOLD,25));
        add(heading);

        JLabel name = new JLabel("Name");
        name.setBounds(50,150,150,30);
        name.setFont(new Font("SAN_SERIF", Font.BOLD,20));
        add(name);

        tname = new JTextField();
        tname.setBounds(200,150,150,30);
        tname.setBackground(new Color(177,252,197));
        add(tname);
```

```
JLabel fname = new JLabel("Father's Name");
fname.setBounds(400,150,150,30);
fname.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(fname);

tfname = new JTextField();
tfname.setBounds(600,150,150,30);
tfname.setBackground(new Color(177,252,197));
add(tfname);

JLabel dob = new JLabel("Date Of Birth");
dob.setBounds(50,200,150,30);
dob.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(dob);

tdob = new JDateChooser();
tdob.setBounds(200,200,150,30);
tdob.setBackground(new Color(177,252,197));
add(tdob);

JLabel salary = new JLabel("Salary");
salary.setBounds(400,200,150,30);
salary.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(salary);

tsalary = new JTextField();
tsalary.setBounds(600,200,150,30);
tsalary.setBackground(new Color(177,252,197));
add(tsalary);

JLabel address = new JLabel("Address");
address.setBounds(50,250,150,30);
address.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(address);

taddress= new JTextField();
taddress.setBounds(200,250,150,30);
taddress.setBackground(new Color(177,252,197));
add(taddress);


JLabel phone = new JLabel("Phone");
phone.setBounds(400,250,150,30);
phone.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(phone);

tphone= new JTextField();
tphone.setBounds(600,250,150,30);
tphone.setBackground(new Color(177,252,197));
add(tphone);
```

```
        JLabel email = new JLabel("Email");
        email.setBounds(50,300,150,30);
        email.setFont(new Font("SAN_SERIF", Font.BOLD,20));
        add(email);

        temail= new JTextField();
        temail.setBounds(200,300,150,30);
        temail.setBackground(new Color(177,252,197));
        add(temail);

        JLabel education = new JLabel("Higest Education");
        education.setBounds(400,300,150,30);
        education.setFont(new Font("SAN_SERIF", Font.BOLD,20));
        add(education);

        String items[] = {"BBA","B.Tech","BCA", "BA", "BSC", "B.COM", "MBA", "MCA", "MA",
"MTech", "MSC", "PHD"};
        Boxeducation = new JComboBox(items);
        Boxeducation.setBackground(new Color(177,252,197));
        Boxeducation.setBounds(600,300,150,30);
        add(Boxeducation);

        JLabel aadhar = new JLabel("Aadhar Number");
        aadhar.setBounds(400,350,150,30);
        aadhar.setFont(new Font("SAN_SERIF", Font.BOLD,20));
        add(aadhar);

        taadhar= new JTextField();
        taadhar.setBounds(600,350,150,30);
        taadhar.setBackground(new Color(177,252,197));
        add(taadhar);


        JLabel empid = new JLabel("Employee ID");
        empid.setBounds(50,400,150,30);
        empid.setFont(new Font("SAN_SERIF", Font.BOLD,20));
        add(empid);

        tempid= new JLabel(""+number);
        tempid.setBounds(200,400,150,30);
        tempid.setFont(new Font("SAN_SARIF", Font.BOLD,20));
        tempid.setForeground(Color.RED);
        add(tempid);

        JLabel designation = new JLabel("Designation");
        designation.setBounds(50,350,150,30);
        designation.setFont(new Font("SAN_SERIF", Font.BOLD,20));
        add(designation);

        tdesignation= new JTextField();
        tdesignation.setBounds(200,350,150,30);
```

```
        tdesignation.setBackground(new Color(177,252,197));
        add(tdesignation);

        add = new JButton("ADD");
        add.setBounds(450,550,150,40);
        add.setBackground(Color.black);
        add.setForeground(Color.WHITE);
        add.addActionListener(this);
        add(add);

        back = new JButton("BACK");
        back.setBounds(250,550,150,40);
        back.setBackground(Color.black);
        back.setForeground(Color.WHITE);
        back.addActionListener(this);
        add(back);


        setSize(900,700);
        setLayout(null);
        setLocation(300,50);
        setVisible(true);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == add){
            String name = tname.getText();
            String fname = tfname.getText();
            String dob = ((JTextField) tdob.getDateEditor().getUiComponent()).getText();
            String salary = tsalary.getText();
            String address = taddress.getText();
            String aadhar = taadhar.getText();
            String phone = tphone.getText();
            String email = temail.getText();
            String education = (String) Boxeducation.getSelectedItem();
            String designation = tdesignation.getText();
            String empID = tempid.getText();

            try{
                conn c = new conn();
                String query = "insert into employee values('"+name+"', '"+fname+"', '"+dob+"',
'"+salary+"','"+address+"', '"+phone+"', '"+email+"', '"+education+"', '"+designation+"','"+aadhar+"',
'"+empID+"')";
                c.statement.executeUpdate(query);
                JOptionPane.showMessageDialog(null,"Details added successfully");
                setVisible(false);
                new Main_class();

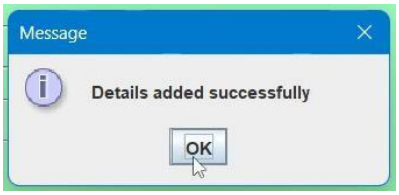
            }catch (Exception E){
                E.printStackTrace();
```

```
            }

        }else {
            setVisible(false);
            new Main_class();
        }
    }

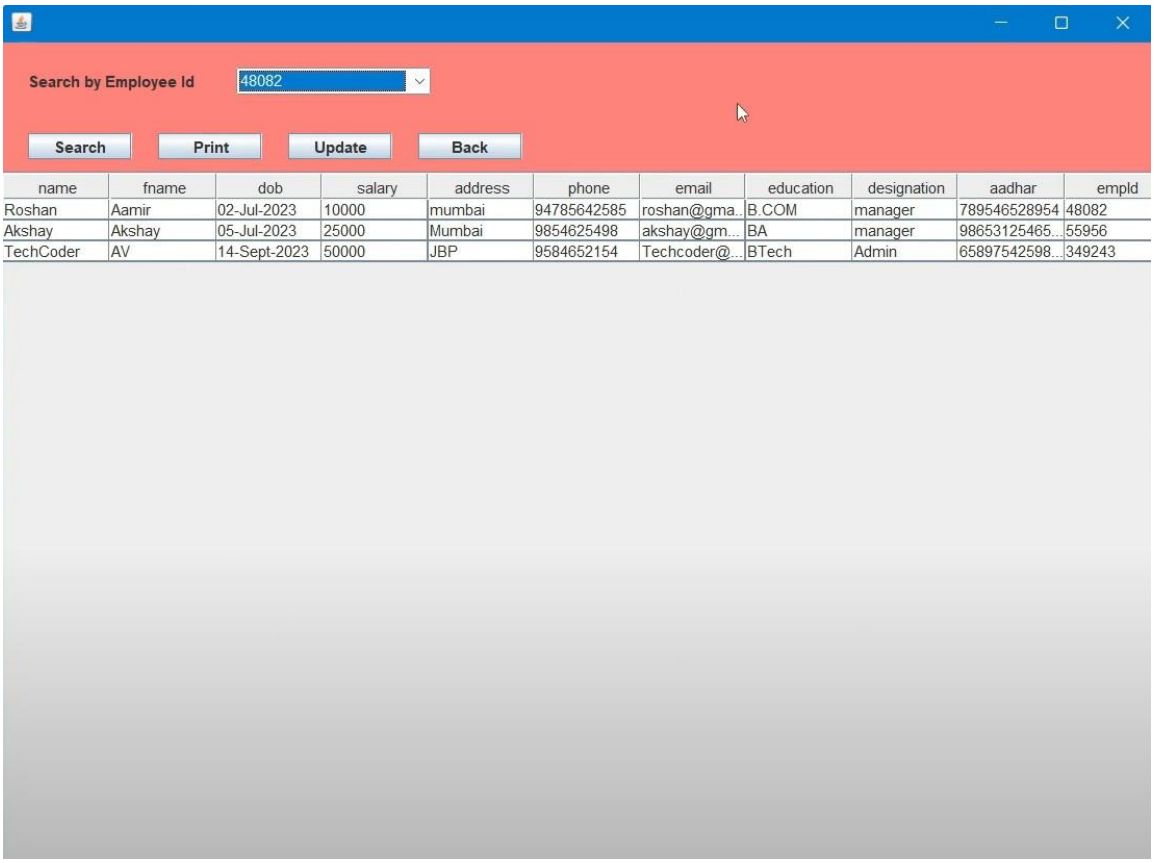
    public static void main(String[] args) {
        new AddEmployee();

    }
}
```



5.4 VIEW EMPLOYEE:





name	fname	dob	salary	address	phone	email	education	designation	aadhar	empId
Roshan	Aamir	02-Jul-2023	10000	mumbai	94785642585	roshan@gma...	B.COM	manager	789546528954	48082
Akshay	Akshay	05-Jul-2023	25000	Mumbai	9854625498	akshay@gm...	BA	manager	98653125465...	55956
TechCoder	AV	14-Sept-2023	50000	JBP	9584652154	Techcoder@...	BTech	Admin	65897542598...	349243

```
package employee.management.system;

import net.proteanit.sql.DbUtils;

import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.sql.ResultSet;

public class View_Employee extends JFrame implements ActionListener {

    JTable table;
    Choice choiceEMP;
    JButton searchbtn, print, update, back;
    View_Employee(){

        getContentPane().setBackground(new Color(255,131,122));
        JLabel search = new JLabel("Search by employee id");
        search.setBounds(20,20,150,20);
        add(search);

        choiceEMP = new Choice();
        choiceEMP.setBounds(180,20,150,20);
        add(choiceEMP);

        try{
            conn c = new conn();
            ResultSet resultSet = c.statement.executeQuery("select * from employee");
            while (resultSet.next()){
                choiceEMP.add(resultSet.getString("empId"));
            }
        }catch (Exception e){
            e.printStackTrace();
        }

        table = new JTable();
        try{
            conn c= new conn();
            ResultSet resultSet = c.statement.executeQuery("select * from employee");
            table.setModel(DbUtils.resultSetToTableModel(resultSet));
        }catch (Exception e){
            e.printStackTrace();
        }

        JScrollPane jp = new JScrollPane(table);
        jp.setBounds(0,100,900,600);
        add(jp);

        searchbtn = new JButton("Search");
        searchbtn.setBounds(20,70,80,20);
```

```
    searchbtn.addActionListener(this);
    add(searchbtn);

    print = new JButton("Print");
    print.setBounds(120,70,80,20);
    print.addActionListener(this);
    add(print);

    update = new JButton("Update");
    update.setBounds(220,70,80,20);
    update.addActionListener(this);
    add(update);

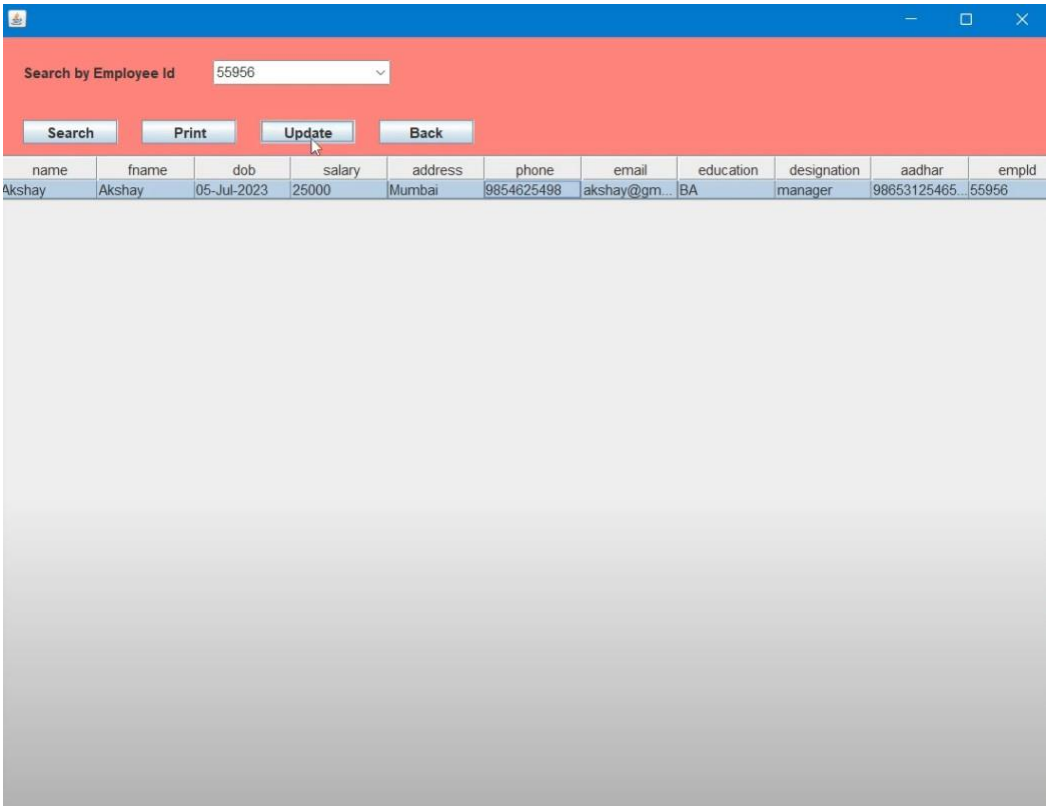
    back = new JButton("Back");
    back.setBounds(320,70,80,20);
    back.addActionListener(this);
    add(back);

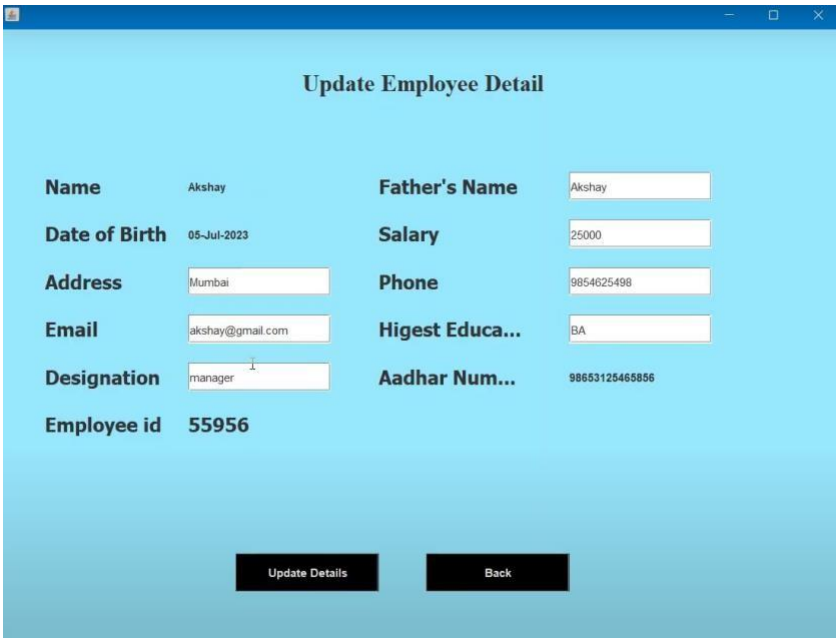

            setSize(900,700);
            setLayout(null);
            setLocation(300,100);
            setVisible(true);
}

@Override
public void actionPerformed(ActionEvent e) {
    if (e.getSource() == searchbtn){
        String query = "select * from employee where empId = '"+choiceEMP.getSelectedItem()+"'";
        try {
            conn c = new conn();
            ResultSet resultSet = c.statement.executeQuery(query);
            table.setModel(DbUtils.resultSetToTableModel(resultSet));
        }catch (Exception E){
            E.printStackTrace();
        }
    } else if (e.getSource() == print) {
        try {
            table.print();
        }catch (Exception E){
            E.printStackTrace();
        }
    } else if (e.getSource() == update){
        setVisible(false);
        new UpdateEmployee(choiceEMP.getSelectedItem());
    } else {
        setVisible(false);
        new Main_class();
    }
}
```

```
    public static void main(String[] args) {
        new View_Employee();
    }
}
```

5.5 UPDATE EMPLOYEE:







```
package employee.management.system;

import com.sun.tools.javac.Main;
import com.toedter.calendar.JDateChooser;

import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.sql.ResultSet;

public class UpdateEmployee extends JFrame implements ActionListener {
    JTextField teducation, tfname, taddress,tphone, taadhar, temail, tsalary,tdesignation;
    JLabel tempid;
    JButton add,back;
    String number;
    UpdateEmployee(String number){

        this.number = number;
        getContentPane().setBackground(new Color(163,255,188));

        JLabel heading = new JLabel("Add Employee Detail");
        heading.setBounds(320,30,500,50);
        heading.setFont(new Font("serif", Font.BOLD,25));
        add(heading);

        JLabel name = new JLabel("Name");
        name.setBounds(50,150,150,30);
        name.setFont(new Font("SAN_SERIF", Font.BOLD,20));
        add(name);

        JLabel tname = new JLabel();
        tname.setBounds(200,150,150,30);
        tname.setBackground(new Color(177,252,197));
        add(tname);

        JLabel fname = new JLabel("Father's Name");
        fname.setBounds(400,150,150,30);
        fname.setFont(new Font("SAN_SERIF", Font.BOLD,20));
        add(fname);

        tfname = new JTextField();
        tfname.setBounds(600,150,150,30);
        tfname.setBackground(new Color(177,252,197));
        add(tfname);

        JLabel dob = new JLabel("Date Of Birth");
        dob.setBounds(50,200,150,30);
        dob.setFont(new Font("SAN_SERIF", Font.BOLD,20));
        add(dob);
```

```
JLabel tdob = new JLabel();
tdob.setBounds(200,200,150,30);
tdob.setFont(new Font("Tahoma", Font.BOLD, 20));
add(tdob);

JLabel salary = new JLabel("Salary");
salary.setBounds(400,200,150,30);
salary.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(salary);

tsalary = new JTextField();
tsalary.setBounds(600,200,150,30);
tsalary.setBackground(new Color(177,252,197));
add(tsalary);

JLabel address = new JLabel("Address");
address.setBounds(50,250,150,30);
address.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(address);

taddress= new JTextField();
taddress.setBounds(200,250,150,30);
taddress.setBackground(new Color(177,252,197));
add(taddress);


JLabel phone = new JLabel("Phone");
phone.setBounds(400,250,150,30);
phone.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(phone);

tphone= new JTextField();
tphone.setBounds(600,250,150,30);
tphone.setBackground(new Color(177,252,197));
add(tphone);

JLabel email = new JLabel("Email");
email.setBounds(50,300,150,30);
email.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(email);

temail= new JTextField();
temail.setBounds(200,300,150,30);
temail.setBackground(new Color(177,252,197));
add(temail);

JLabel education = new JLabel("Higest Education");
education.setBounds(400,300,150,30);
education.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(education);
```

```
teducation= new JTextField();
teducation.setBounds(600,300,150,30);
teducation.setBackground(new Color(177,252,197));
add(teducation);

JLabel aadhar = new JLabel("Aadhar Number");
aadhar.setBounds(400,350,150,30);
aadhar.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(aadhar);

JLabel taadhar= new JLabel();
taadhar.setBounds(600,350,150,30);
taadhar.setBackground(new Color(177,252,197));
add(taadhar);


JLabel empid = new JLabel("Employee ID");
empid.setBounds(50,400,150,30);
empid.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(empid);

tempid= new JLabel();
tempid.setBounds(200,400,150,30);
tempid.setFont(new Font("SAN_SARIF", Font.BOLD,20));
tempid.setForeground(Color.RED);
add(tempid);

JLabel designation = new JLabel("Designation");
designation.setBounds(50,350,150,30);
designation.setFont(new Font("SAN_SERIF", Font.BOLD,20));
add(designation);

tdesignation= new JTextField();
tdesignation.setBounds(200,350,150,30);
tdesignation.setBackground(new Color(177,252,197));
add(tdesignation);

try {
    conn c = new conn();
    String query = "select * from employee where empId = '"+number+"'";
    ResultSet resultSet = c.statement.executeQuery(query);
    while (resultSet.next()){
        tname.setText(resultSet.getString("name"));
        tfname.setText(resultSet.getString("fname"));
        tdob.setText(resultSet.getString("dob"));
        taddress.setText(resultSet.getString("address"));
        tsalary.setText(resultSet.getString("salary"));
        tphone.setText(resultSet.getString("phone"));
        temail.setText(resultSet.getString("email"));
        teducation.setText(resultSet.getString("education"));
        taadhar.setText(resultSet.getString("addhar"));
```

```
            tempid.setText(resultSet.getString("empId"));
            tdesignation.setText(resultSet.getString("designation"));
          }
        }catch (Exception e){
          e.printStackTrace();
        }


        add = new JButton("UPDATE");
        add.setBounds(450,550,150,40);
        add.setBackground(Color.black);
        add.setForeground(Color.WHITE);
        add.addActionListener(this);
        add(add);

        back = new JButton("BACK");
        back.setBounds(250,550,150,40);
        back.setBackground(Color.black);
        back.setForeground(Color.WHITE);
        back.addActionListener(this);
        add(back);


        setSize(900,700);
        setLayout(null);
        setLocation(300,50);
        setVisible(true);

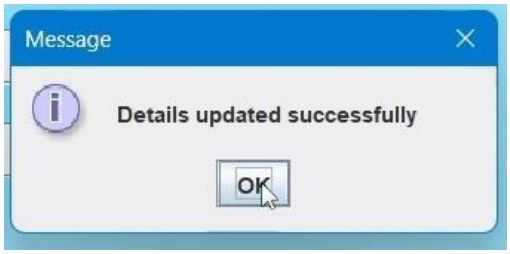
    }

    @Override
    public void actionPerformed(ActionEvent e) {
      if (e.getSource() == add){
          String fname = tfname.getText();
          String salary = tsalary.getText();
          String address = taddress.getText();
          String phone = tphone.getText();
          String email = temail.getText();
          String education = teducation.getText();
          String designation = tdesignation.getText();

          try {
              conn c = new conn();
              String query = "update employee set fname = '"+fname+"', salary = '"+salary+"', address =
'"+address+"', phone = '"+phone+"', email = '"+email+"', education = '"+education+"', designation =
'"+designation+"' where empId = '"+number+"'";
              c.statement.executeUpdate(query);
              JOptionPane.showMessageDialog(null, "Details updated successfully");
              setVisible(false);
              new Main_class();
          }catch (Exception E){
```

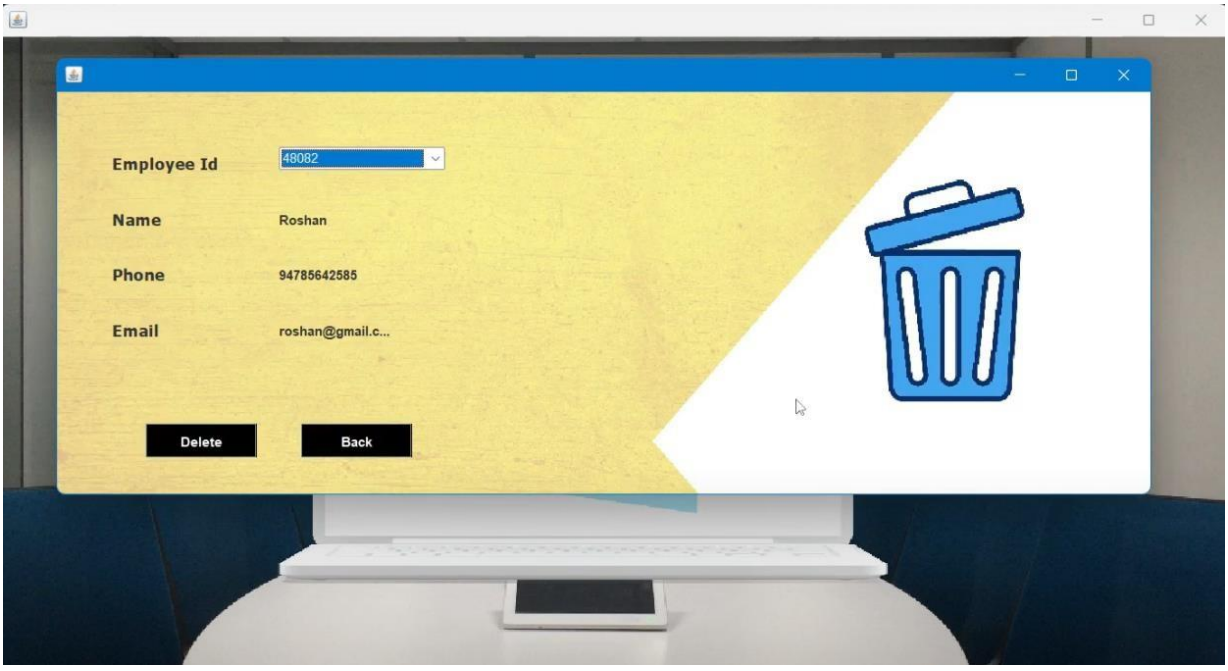
```
            E.printStackTrace();
        }
      }else {
        setVisible(false);
        new View_Employee();
      }
    }

    public static void main(String[] args) {
      new UpdateEmployee("");
    }
}
```



5.6 REMOVE EMPLOYEE:





```
package employee.management.system;

import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.awt.event.ItemEvent;
import java.awt.event.ItemListener;
import java.sql.ResultSet;

public class RemoveEmployee extends JFrame implements ActionListener {
    Choice choiceEMPID;
    JButton delete, back;
    RemoveEmployee(){

        JLabel label = new JLabel("Employee ID");
        label.setBounds(50,50,100,30);
        label.setFont(new Font("Tahoma", Font.BOLD,15));
        add(label);

        choiceEMPID = new Choice();
        choiceEMPID.setBounds(200,50,150,30);
        add(choiceEMPID);

        try{
            conn c = new conn();
            ResultSet resultSet = c.statement.executeQuery("select * from employee");
            while (resultSet.next()){
                choiceEMPID.add(resultSet.getString("empId"));
            }
        }catch (Exception e){
            e.printStackTrace();
        }

        JLabel labelName = new JLabel("Name");
        labelName.setBounds(50,100,100,30);
        labelName.setFont(new Font("Tahoma", Font.BOLD,15));
        add(labelName);

        JLabel textName = new JLabel();
        textName.setBounds(200,100,100,30);
        add(textName);

        JLabel labelPhone = new JLabel("Phone");
        labelPhone.setBounds(50,150,100,30);
        labelPhone.setFont(new Font("Tahoma", Font.BOLD,15));
        add(labelPhone);

        JLabel textPhone = new JLabel();
        textPhone.setBounds(200,150,100,30);
        add(textPhone);
```

```
        JLabel labelemail = new JLabel("Email");
        labelemail.setBounds(50,200,100,30);
        labelemail.setFont(new Font("Tahoma", Font.BOLD,15));
        add(labelemail);

        JLabel textEmail = new JLabel();
        textEmail.setBounds(200,200,100,30);
        add(textEmail);

        try {
            conn c = new conn();
            ResultSet resultSet = c.statement.executeQuery("select * from employee where empId =
'"+choiceEMPID.getSelectedItem()+"'");
            while (resultSet.next()){
                textName.setText(resultSet.getString("name"));
                textPhone.setText(resultSet.getString("phone"));
                textEmail.setText(resultSet.getString("email"));
            }
        }catch (Exception e){
            e.printStackTrace();
        }

        choiceEMPID.addItemListener(new ItemListener() {
            @Override
            public void itemStateChanged(ItemEvent e) {
                try{
                    conn c = new conn();
                    ResultSet resultSet = c.statement.executeQuery("select * from employee where empId =
'"+choiceEMPID.getSelectedItem()+"'");
                    while (resultSet.next()) {
                        textName.setText(resultSet.getString("name"));
                        textPhone.setText(resultSet.getString("phone"));
                        textEmail.setText(resultSet.getString("email"));
                    }
                }catch (Exception E){
                    E.printStackTrace();
                }
            }
        });

        delete = new JButton("Delete");
        delete.setBounds(80,300,100,30);
        delete.setBackground(Color.black);
        delete.setForeground(Color.WHITE);
        delete.addActionListener(this);
        add(delete);

        back = new JButton("Back");
        back.setBounds(220,300,100,30);
        back.setBackground(Color.black);
```

```
        back.setForeground(Color.WHITE);
        back.addActionListener(this);
        add(back);

        ImageIcon i1 = new ImageIcon(ClassLoader.getSystemResource("icons/delete.png"));
        Image i2 = i1.getImage().getScaledInstance(200,200,Image.SCALE_DEFAULT);
        ImageIcon i3 = new ImageIcon(i2);
        JLabel img = new JLabel(i3);
        img.setBounds(700,80,200,200);
        add(img);

        ImageIcon i11 = new ImageIcon(ClassLoader.getSystemResource("icons/rback.png"));
        Image i22 = i11.getImage().getScaledInstance(1120,630,Image.SCALE_DEFAULT);
        ImageIcon i33 = new ImageIcon(i22);
        JLabel image = new JLabel(i33);
        image.setBounds(0,0,1120,630);
        add(image);

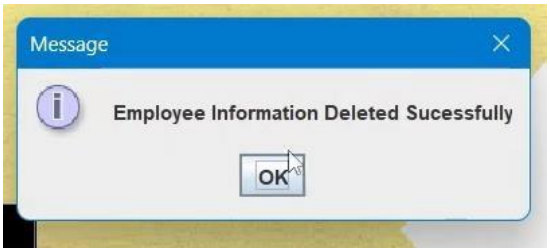
        setSize(1000,400);
        setLocation(300,150);
        setLayout(null);
        setVisible(true);

    }

    @Override
    public void actionPerformed(ActionEvent e) {
        if (e.getSource()==delete){
            try {
                conn c = new conn();
                String query = "delete from employee where empId = '"+choiceEMPID.getSelectedItem()+"'";
                c.statement.executeUpdate(query);
                JOptionPane.showMessageDialog(null,"Employee Deleted Sucessfully");
                setVisible(false);
                new Main_class();

            }catch (Exception E){
                E.printStackTrace();
            }
        }else {
            setVisible(false);
        }
    }

    public static void main(String[] args) {
        new RemoveEmployee();
    }
}
```


Message
Employee Information Deleted Sucessfully
OK

6 - CONCLUSION

In conclusion, our employee database management system implemented in Java using JDBC successfully addresses the core functionalities of adding new employee details, removing records, and efficiently retrieving data when needed. The utilization of JDBC allowed seamless interaction between the Java application and the database, ensuring robust data management.

The project emphasized the importance of adopting design patterns, such as the Singleton pattern, to enhance code structure and maintainability. Additionally, the project's reliance on established database management principles, as outlined in key texts, ensured a solid foundation for data integrity and security.

Through this project, we gained valuable insights into Java application development, database connectivity, and the significance of adhering to best practices in software design. The system provides a user-friendly interface for managing employee records, contributing to a more efficient and organized human resources management process.

7 - REFERENCES

JAVA- The complete reference by ORACLE

YOUTUBE (LINK BELOW)

https://www.youtube.com/playlist?list=PLbKub4Jss9oUbTmeS9li_DoskYCiuvx1R